

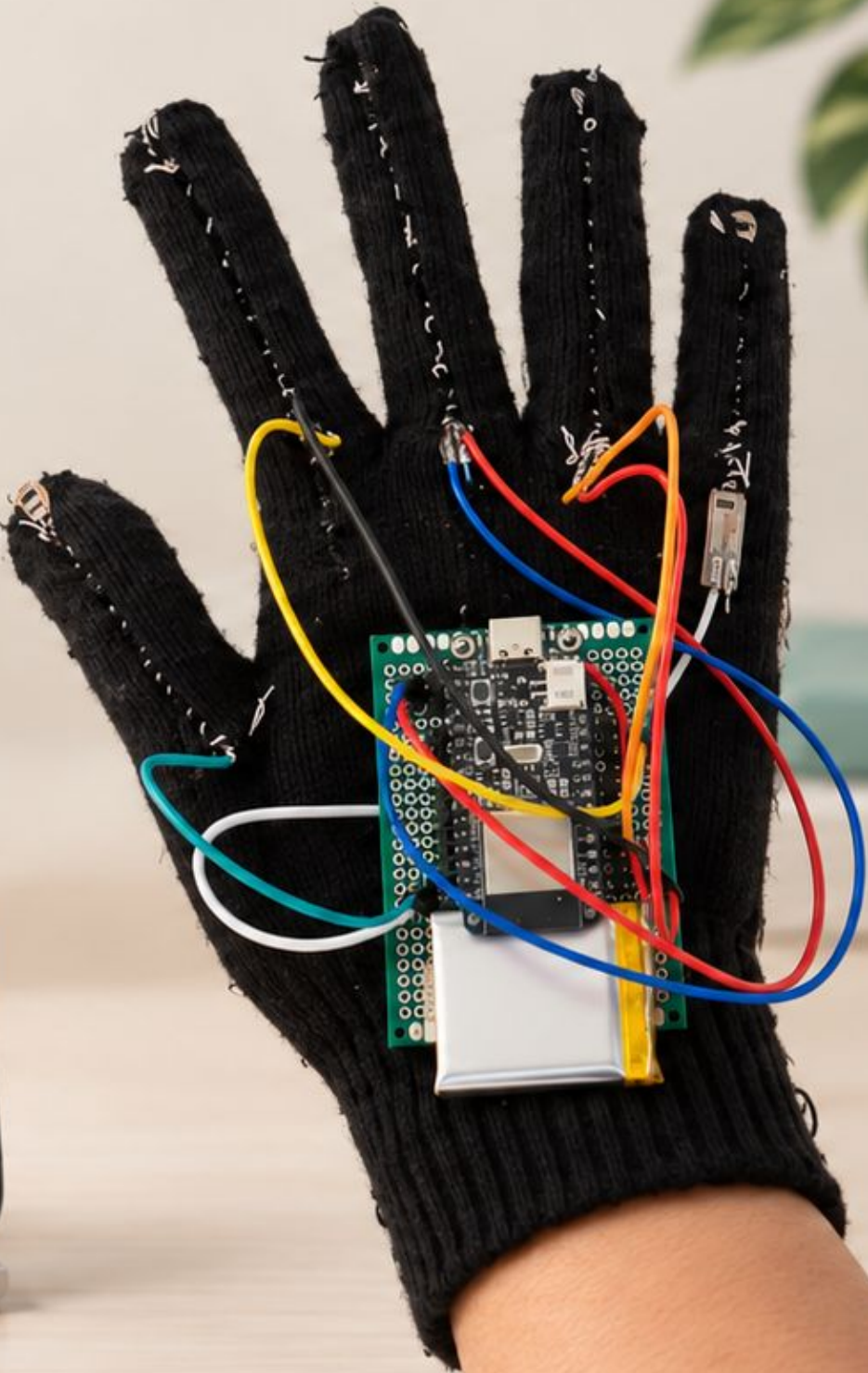
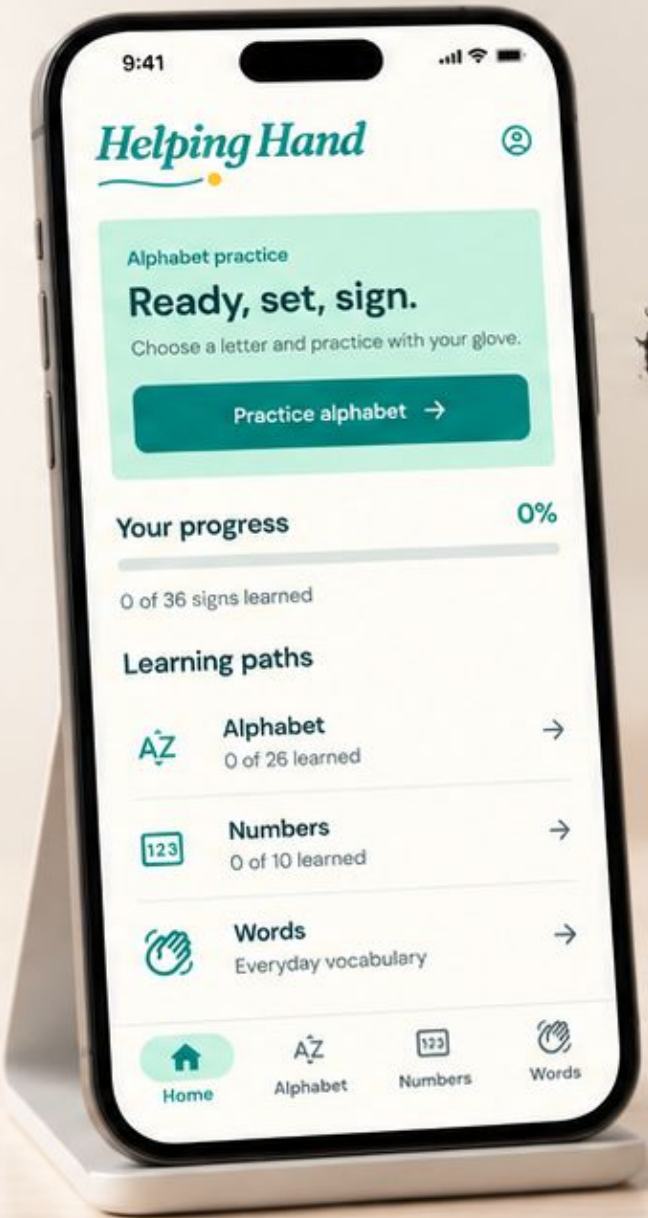
BETA BUILD · OCTOBER 6, 2026

Helping Hand

A Wearable ASL Learning System

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THE PROBLEM

Learning ASL needs feedback.

>90%

of deaf children are born to hearing parents.

Books and videos show the sign.

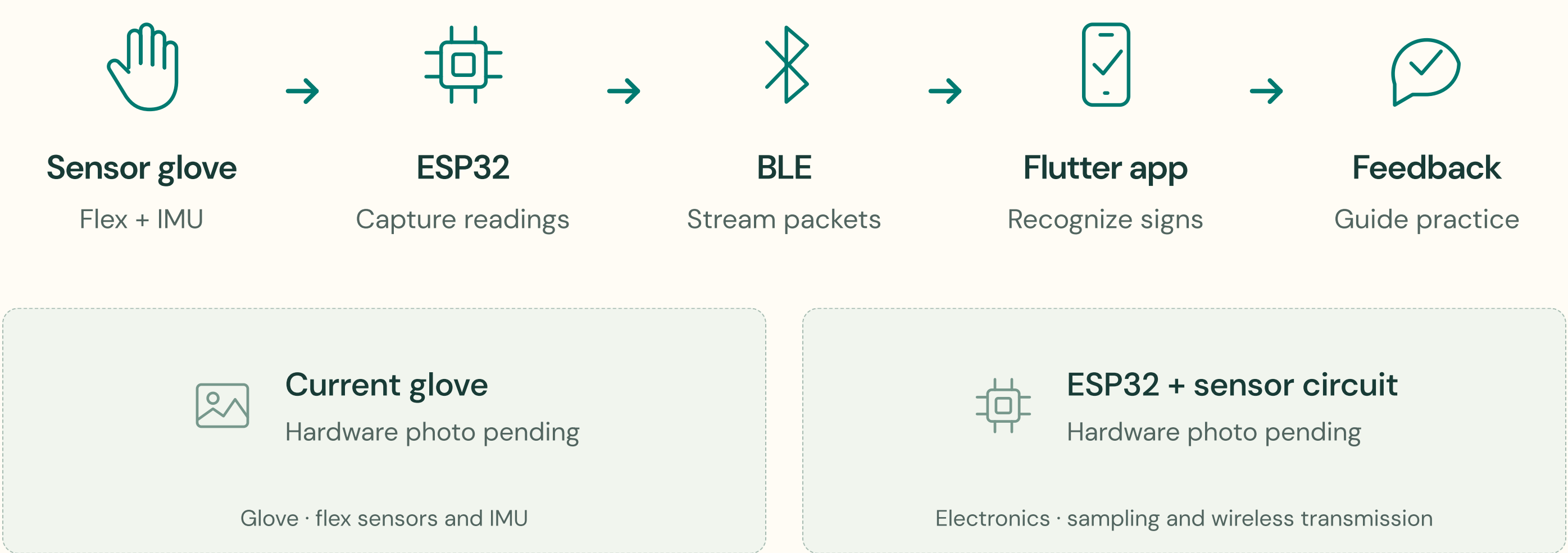
2D examples cannot assess your hand position.

No live correction from the reference alone.



OUR SOLUTION

Wear. Sign. Get feedback.



The first complete pipeline



Alpha glove



Alpha interface

- 01

Sensor capture

Finger bend + motion readings
- 02

Wireless streaming

Glove → BLE → app
- 03

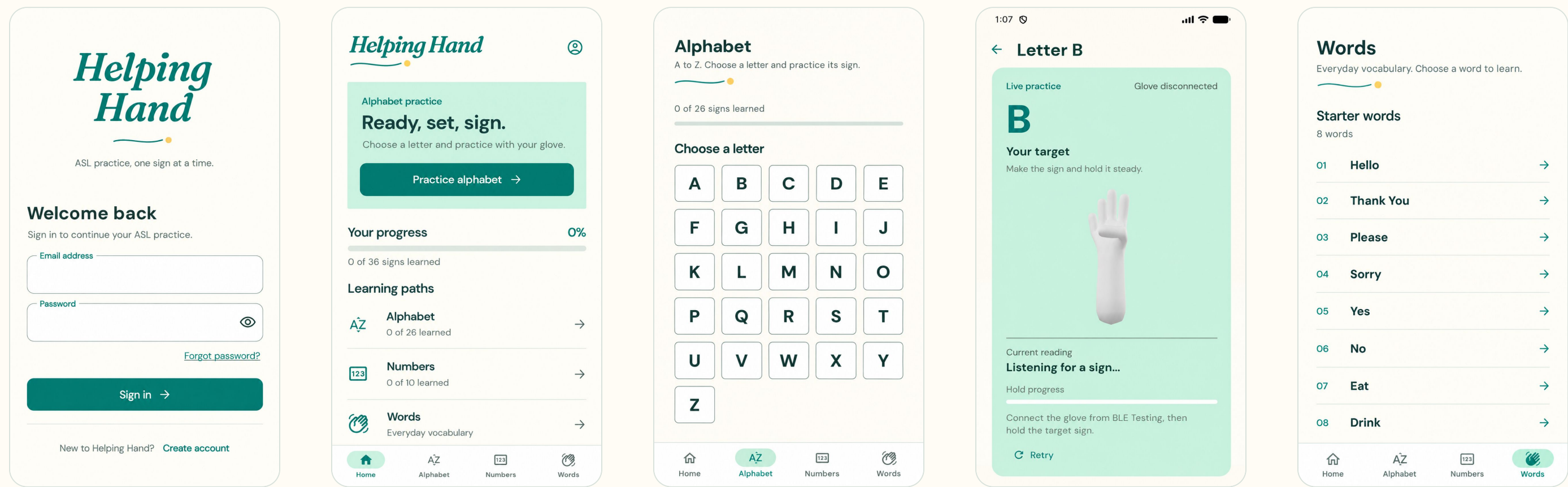
Static classification

36-class TFLite baseline

THE BETA APP

A clearer path to practice

Firebase sign-in · dedicated practice pages · account-specific progress



01 Sign in

02 Choose a path

03 Pick a target

04 Practice

05 Explore words

Recognizing words with a TCN

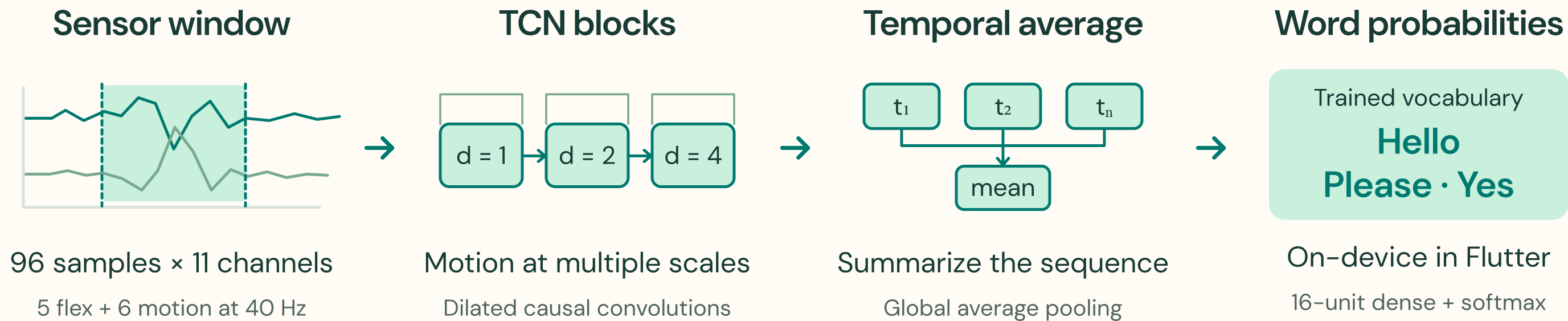
Static · MLP

One sensor snapshot → static class

Words · Temporal Convolutional Network

Sensor window → word prediction

Resample, trim or pad, and standardize before inference



Compared with **CNN-GRU** and **CNN-LSTM**

Selected export · **48.3 kB TFLite**

HELPING HAND IN ACTION

LIVE DEMO!!!

Static sign → Word attempts

Implementation and remaining validation

Completed implementation

Core experience

- Static MLP + 3-word TCN
- Firebase accounts + lessons
- Saved sign + word progress

In progress

Live validation

- Glove + phone rehearsal
- Rest / unknown rejection
- Wrist angles + other wearers

Planned

Tactile guidance

- 10 planned piezo elements
- Error-to-vibration mapping
- Calibration + user testing

Static completion rules

≥80%

Confidence

≥750 ms

Stable matching hold

≥5

Matching packets

≤250 ms

Gap between packets

NEXT MILESTONE

Feel what needs correcting.

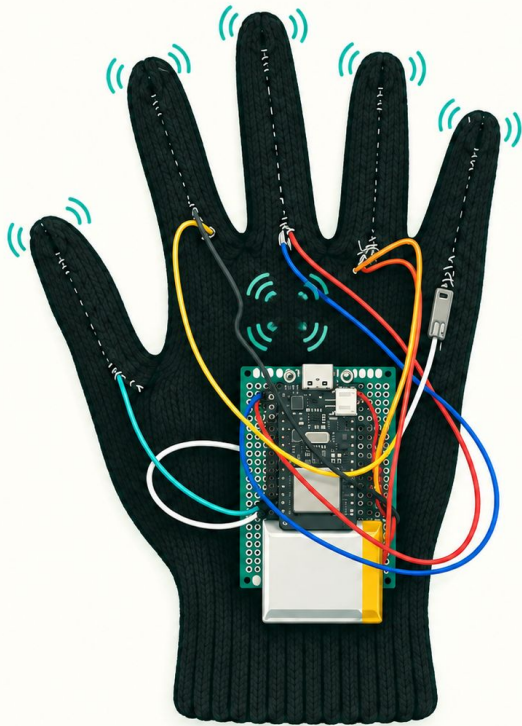
5 + 5

X pattern + one on each finger

Planned piezo feedback for incorrect positioning.

Estimated deviation → Pulse pattern / intensity

Placement, error mapping, and tactile response need calibration.



Concept illustration · planned feature

●	●	●	●
Oct 7–13	Oct 14–20	Oct 21–27	Oct 28–Nov 3
Mount + wire	Calibrate guidance	Refine recognition	Test + polish
Actuators and drivers	Error → tactile response	Orientation + signer trials	User trials and release prep

Helping Hand

Thank you. Questions?

Practice with feedback.

Next: guidance you can feel.